

STEM MENTORSHIP PROGRAMME

STEM High School for Boys 6th of October
STEM Mentorship

ENGLISH

English MASTERY GUIDE

Comprehensive English Language Reference for Academic Excellence

A Complete Resource for Grammar, Vocabulary, and Communication Skills

2026 Edition

Excellence · Leadership · Innovation · Integrity

This comprehensive guide is designed specifically for ambitious STEM students preparing for entrance examinations and academic excellence. Through detailed explanations, practical examples, and strategic insights, this resource will strengthen your English language proficiency and communication abilities essential for success in a STEM education environment.

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Chapter I

Introduction and Guidance

1 Welcome to STEM English Learning

Welcome to the STEM English Mastery Guide—a comprehensive resource designed specifically for students pursuing excellence in STEM education. This guide recognizes that strong English language proficiency is fundamental to success in STEM fields, where clear communication, precise technical writing, and accurate comprehension are essential.

Whether you are preparing for entrance examinations, strengthening your academic writing, or simply committed to mastering the English language, this guide provides the tools, insights, and strategies needed to achieve your goals.

1.1 Why English Matters in STEM

In STEM fields, English proficiency directly impacts your ability to:

- **Understand complex scientific and technical texts** with precision and depth
- **Communicate findings clearly** through written reports, research papers, and presentations
- **Collaborate effectively** with peers and mentors using proper technical terminology
- **Access global resources** published primarily in English
- **Present ideas persuasively** to audiences ranging from classrooms to professional conferences
- **Write compelling applications** for universities, scholarships, and opportunities

1.2 What This Guide Covers

This comprehensive resource includes:

1. **13 Essential Verb Tenses** with usage rules, structures, and real-world examples
2. **Complete Pronoun Reference** including subject, object, and possessive forms
3. **All Eight Parts of Speech** with detailed explanations and practical applications

4. **Passive and Active Voice** with transformation techniques and contextual usage
5. **Five Types of Conditional Sentences** with real-world STEM examples
6. **Advanced Conjunctions** organized by function and purpose
7. **116+ Phrasal Verbs** essential for natural English communication
8. **Word Formation Rules** including prefixes, suffixes, and compound structures
9. **20+ Common Idioms** with context and appropriate usage
10. **Complete Vocabulary Reference** with antonyms, synonyms, and homonyms
11. **Advanced Topics** including relative clauses, countable/uncountable nouns, and context clues
12. **Practical Application Strategies** for writing, speaking, and communication in academic settings

2 How to Maximize This Guide

2.1 For Examination Preparation

If you are preparing for entrance examinations:

1. **Begin with Diagnostics:** Assess which areas need improvement using the self-assessment checklist
2. **Study Systematically:** Progress through each chapter sequentially to build comprehensive understanding
3. **Focus on Weak Areas:** Spend additional time on sections where you struggle most
4. **Memorize Key Structures:** Pay special attention to verb tense formations and exception rules
5. **Practice Actively:** Apply concepts to sample sentences and create your own examples
6. **Review Regularly:** Revisit challenging topics multiple times before examinations
7. **Use the Quick Reference:** Before exams, review the quick reference section to refresh your memory

2.2 For Academic Writing Improvement

To strengthen your academic writing:

1. **Study Chapter 3:** Master passive voice and complex sentences essential for formal writing
2. **Focus on Chapter 4:** Build vocabulary through idioms, synonyms, and word families
3. **Review Chapter 7:** Implement transitional phrases and academic writing techniques
4. **Practice Writing:** Apply these concepts by writing essays, reports, and analyses
5. **Edit Carefully:** Review your writing for grammatical accuracy and clarity
6. **Study Examples:** Analyze well-written academic texts to recognize professional structures

2.3 For Daily English Improvement

To continuously strengthen your English:

1. **Read Strategically:** Engage with academic articles, scientific journals, and quality publications
2. **Study Vocabulary:** Learn 5-10 new words daily using the vocabulary sections
3. **Practice Speaking:** Apply phrasal verbs and expressions in conversation
4. **Write Consistently:** Maintain a journal or write practice essays
5. **Consume English Media:** Watch documentaries, TED talks, and educational videos
6. **Reflect on Mistakes:** When you make errors, identify the rule and review it thoroughly

3 Study Strategies for Academic Success

3.1 The Spaced Repetition Method

Research shows that spaced repetition significantly improves long-term retention:

- **Day 1:** Study a new concept thoroughly
- **Day 3:** Review the same concept
- **Day 7:** Review again after one week
- **Day 14:** Review after two weeks
- **Day 30:** Final review before examination

3.2 Active Learning Techniques

Instead of passive reading, engage actively:

1. **Create Flashcards:** Make physical or digital flashcards for difficult concepts
2. **Write Practice Sentences:** For each grammar rule, write 5-10 example sentences
3. **Explain Aloud:** Teach concepts to others or speak them aloud to yourself
4. **Make Connections:** Link new vocabulary to words you already know
5. **Create Mind Maps:** Visualize relationships between concepts
6. **Teach Others:** Explaining to friends reinforces your own understanding

Chapter II

Foundational Grammar: Building Your Language Foundation

1 Verb Tenses and Sentence Structures

Verbs are the foundation of English sentences. Understanding the complete tense system is absolutely essential for grammatical accuracy and natural communication.

1.1 Overview of the Complete Tense System

English has 13 main tense forms organized around three time periods: past, present, and future. Each time period has four aspects: simple, perfect, continuous, and perfect continuous.

Time Period	Simple	Continuous	Perfect
Present	Present Simple	Pres. Continuous	Pres. Perfect
Past	Past Simple	Past Continuous	Past Perfect
Future	Future Simple	Future Continuous	Future Perfect
Perfect Continuous		Present Perfect Continuous	–

1.2 1. Simple Present Tense

Structure: Subject + Base Verb + Object/Complement

Formation Rules:

- Third person singular adds -s or -es: “He *plays*”, “She *watches*”
- Negative form: does not / doesn’t + base verb
- Question form: Do/Does + subject + base verb

Primary Uses:

1. **Habitual Actions:** “I exercise every morning.”
2. **General Facts:** “Water boils at 100°C.”
3. **Permanent States:** “She lives in Cairo.”
4. **Scientific Truths:** “Light travels faster than sound.”
5. **Schedules:** “The meeting starts at 3 PM.”

6. **Instructions:** “You mix the chemicals carefully.”

Time Keywords: always, usually, often, sometimes, seldom, never, daily, weekly, monthly, every day, on Saturdays

Key Insight: Simple present describes habitual actions or permanent truths, NOT actions happening right now. For current actions, you must use present continuous.

Common Mistake: “I am knowing the answer” (INCORRECT). Correct: “I know the answer.” (Stative verbs don’t use continuous forms)

Stative Verbs (Never Used in Continuous): be, have, know, like, love, hate, want, need, prefer, agree, believe, understand, realize, seem, appear, belong, own, possess

1.3 2. Present Continuous

Structure: Subject + am/is/are + Present Participle (-ing)

Examples:

- “I am studying chemistry.”
- “She is conducting an experiment.”
- “They are building a prototype.”

Primary Uses:

1. **Current Action (Right Now):** “I am writing this guide.”
2. **Temporary Situation:** “I am taking extra classes this semester.”
3. **Future Plans:** “We are visiting the university next week.”
4. **Change in Progress:** “Your English is improving significantly.”

Time Keywords: now, at the moment, currently, right now, at present, today, this week, this year

1.4 3. Simple Past

Structure: Subject + Past Verb Form + Object/Complement

Formation Rules:

- Regular verbs add -ed: play → played, study → studied
- Irregular verbs follow unique patterns: go → went, see → saw, have → had
- Negative: did not / didn’t + base verb
- Questions: Did + subject + base verb

Primary Uses:

1. **Completed Actions:** “I finished the report yesterday.”
2. **Sequential Past Events:** “I entered the room, sat down, and opened my laptop.”
3. **Past Habits:** “When I was younger, I played football every Saturday.”
4. **Past States:** “She was tired after the exam.”

Time Keywords: yesterday, last week, last month, in 2023, three days ago, previously, once, then, at that time

1.5 4. Past Continuous

Structure: Subject + was/were + Present Participle (-ing)

Uses:

1. **Action Interrupted by Another:** “I was studying when he called.”
2. **Background Action:** “While she was giving the presentation, I was taking notes.”
3. **Parallel Past Actions:** “They were working on their projects while we were organizing the event.”

1.6 5. Present Perfect

Structure: Subject + have/has + Past Participle

Formation Notes:

- Positive: I/you/we/they have done; he/she/it has done
- Negative: haven't / hasn't done
- Questions: Have/Has + subject + past participle

Critical Uses (Understanding the Connection to Present):

1. **Recent Action with Present Relevance:** “I have just completed the assignment.” (The completion is recent and relevant now)
2. **Experience (Time Unspecified):** “I have visited five countries.” (The experiences are complete, but the exact time is not specified)
3. **Ongoing Situation:** “It has been raining for two hours.” (The rain started in the past and continues now)
4. **Repeated Actions:** “I have taken that exam three times.” (Multiple completions, total is relevant)
5. **Life Accomplishment:** “She has graduated from university.” (A significant life event with present importance)

Time Keywords: already, yet (in negatives/questions), just, ever, never, so far, up to now, recently, for, since, these past months

Critical Distinction:

Present Perfect: “I have written three essays.” (General time, focus on accomplishment)

Simple Past: “I wrote three essays yesterday.” (Specific time, action is complete and finished)

The key difference: Use present perfect when time is NOT specified. Use simple past when you mention WHEN it happened.

1.7 6. Past Perfect

Structure: Subject + had + Past Participle

Primary Use: Showing that one past action happened BEFORE another past action

Example: “I had finished my homework before my friend arrived.”

Explanation: The homework was completed first (past perfect), then the friend arrived (past simple).

Common Pattern in Narratives: “By the time he graduated, he had learned five programming languages.”

1.8 7. Future Simple

Structure: Subject + will + Base Verb

Alternative: Subject + is going to + Base Verb

Uses:

1. **Predictions:** “Artificial intelligence will transform industries.”
2. **Decisions Made Now:** “I will study engineering.”
3. **Promises:** “I will submit the project on time.”
4. **Facts About Future:** “Sunset will occur at 6:30 PM.”

Will vs. Going To:

- Use **will** for spontaneous decisions or predictions
- Use **going to** for planned intentions or near future events

1.9 8-13. Advanced Future and Perfect Continuous Tenses

Future Continuous: “I will be studying all night.” (Activity in progress at a future time)

Future Perfect: “By next year, I will have completed the degree.” (Action completed before a future time)

Present Perfect Continuous: “I have been studying for three hours.” (Action started in past, continues now, emphasis on duration)

Past Perfect Continuous: “I had been waiting for an hour before he arrived.” (Action in progress, then interrupted)

Future Perfect Continuous: “By June, I will have been studying here for two years.” (Duration completed by future time)

2 Pronouns and Possessives

Pronouns replace nouns to avoid repetition and improve clarity. Understanding pronoun types and agreement is essential for clear writing.

2.1 Subject Pronouns (The Actor)

Subject pronouns perform the action in a sentence.

Singular	Plural	Example Sentence
I	We	" I study mathematics every day."
You	You	" You are doing excellent work."
He	They	" He will become a scientist."
She		" She excels in physics."
It		" It functions properly."

2.2 Object Pronouns (The Receiver)

Object pronouns receive the action. They come after verbs or prepositions.

- **After Action Verbs:** "The teacher asked *us* to submit our work."
- **After Prepositions:** "This opportunity is perfect for *him*."
- **Object of Comparison:** "She is taller than *me*."

Singular	Plural	Example
Me	Us	"Help me understand this concept."
You	You	"I congratulate you on your success."
Him	Them	"Give it to him ."
Her		"Tell her about the opportunity."
It		"I appreciate it greatly."

2.3 Possessive Adjectives (Modifying Nouns)

These adjectives show ownership and always come BEFORE a noun.

Possessive Adjective	Example
My	"This is my research project."
Your	"I reviewed your proposal."
His	" His experiment was groundbreaking."
Her	" Her presentation impressed everyone."
Its	"The system has its advantages and disadvantages."
Our	"We completed our assignment."
Their	" Their dedication is admirable."

Critical Note on ITS vs. IT'S:

Its (possessive): "The company released **its** new product." (No apostrophe!)

It's (contraction): "**It's** raining outside." (It is raining)

This is one of the most common mistakes. Remember: **ITS = POSSESSION, IT'S = IT IS**

2.4 Possessive Pronouns (Standing Alone)

Unlike possessive adjectives, possessive pronouns stand alone without a noun following.

Possessive Pronoun	Example
Mine	"This achievement is mine ."
Yours	"Is this laptop yours ?"
His	"The decision was entirely his ."
Hers	"The award belongs to hers ."
Ours	"This responsibility is ours ."
Theirs	"The choice was theirs ."

3 Parts of Speech

Understanding the eight parts of speech is fundamental to analyzing sentence structure and improving your writing.

3.1 The Eight Parts of Speech

Part	Function	Examples	Sentences
Noun	Thing, person, place, or idea	Student, school, chemistry, honesty	" <i>Mathematics</i> is a <i>subject</i> that requires <i>practice</i> ."
Verb	Action or state of being	Run, think, solve, exist, appear	"She <i>conducted</i> the <i>experiment</i> <i>carefully</i> ."
Adjective	Describes a noun	Brilliant, difficult, precise, innovative	"The <i>complex equation</i> requires <i>careful</i> analysis."
Adverb	Modifies verbs, adjectives, or other adverbs	Carefully, frequently, extremely, well	"He solved the problem <i>remarkably</i> quickly."
Pronoun	Replaces a noun	I, you, he, she, it, we, they, who	" <i>She</i> completed her work, and <i>he</i> completed <i>his</i> ."
Preposition	Shows relationship between words	In, on, at, between, during, toward	"The microscope is <i>on</i> the table <i>in</i> the lab."
Conjunction	Joins words, phrases, or clauses	And, but, or, because, although, while	"I studied hard <i>but</i> <i>and</i> yet struggled with the exam."
Interjection	Expresses strong emotion	Oh!, Wow!, Alas!, Bravo!	" <i>Wow!</i> The results are extraordinary!"

3.2 Part-of-Speech Analysis in Context

Consider this sentence: "The brilliant student solved the complex problem quickly."

- **The:** Article (type of adjective)
- **brilliant:** Adjective (describing the student)
- **student:** Noun (person)
- **solved:** Verb (action)
- **the:** Article
- **complex:** Adjective (describing the problem)
- **problem:** Noun (thing)
- **quickly:** Adverb (modifying how he solved)

Chapter III

Advanced Grammatical Structures

1 Passive Voice and Active Voice

Understanding the distinction between active and passive voice is crucial for academic writing and clear communication.

1.1 Understanding the Difference

Active Voice: The subject performs the action.

- “The scientist conducted the experiment.”
- Structure: Subject (Agent) + Verb + Object

Passive Voice: The subject receives the action.

- “The experiment was conducted by the scientist.”
- Structure: Subject (Receiver) + was/were + Past Participle + by + Agent

1.2 Complete Tense Transformation Chart

Active Voice	Passive Voice
does	is done
did	was done
will do	will be done
has done	has been done
had done	had been done
will have done	will have been done
is doing	is being done
was doing	was being done
is being done	is being done

1.3 When to Use Passive Voice

Use Passive Voice When:

1. **The Agent is Unknown:** “The laboratory was broken into.”
2. **The Agent is Unimportant:** “The bridge was completed in 2020.”

3. **Writing Formally:** Scientific reports often use passive voice
4. **Emphasizing the Action:** “The data was carefully analyzed.”
5. **Multiple Agents:** “The project was completed by numerous contributors.”

Scientific Example:

- Active: “We observed that the sample changed color.”
- Passive: “It was observed that the sample changed color.”

2 Conditional Sentences

Conditional sentences express what happens under certain circumstances. Mastering these five types is essential for academic English.

2.1 Type 0: Zero Conditional (Universal Truths)

Structure: If + present simple, present simple

Usage: Universal truths, scientific facts, laws of nature

Examples:

- “If you heat water to 100°C, it boils.”
- “If students study consistently, they improve their grades.”
- “If light enters a prism, it refracts.”

The zero conditional emphasizes that the result is ALWAYS the same when the condition is met.

2.2 Type 1: First Conditional (Probable Future)

Structure: If + present simple, will/can/should + base verb

Usage: Likely future situations, probable outcomes

Examples:

- “If you study hard, you will pass the exam.”
- “If the weather improves, we can conduct the outdoor experiment.”
- “If you submit your work on time, you should receive full credit.”

The first conditional suggests the condition has a reasonable chance of occurring.

2.3 Type 2: Second Conditional (Hypothetical/Unlikely)

Structure: If + past simple, would/could/should + base verb

Usage: Imaginary or unlikely situations

Examples:

- “If I had unlimited resources, I would establish a research institute.”
- “If she were the team leader, she would implement better systems.”
- “If we could reverse climate change, our planet would thrive.”

Note: Use “were” for all subjects with “be” (not “was”)

2.4 Type 3: Third Conditional (Past Impossibility)

Structure: If + past perfect, would have/could have/should have + past participle

Usage: Expressing regret about past situations that cannot be changed

Examples:

- “If I had studied harder, I would have scored higher.” (I didn’t study hard, so I can’t change the score now)
- “If she had known about the opportunity, she would have applied.”
- “If they had invested in technology earlier, they could have saved time.”

2.5 Type 4: Real Conditionals (Mixed)

Structure: If + present simple, past simple (for habitual past results)

Usage: What happened consistently in the past under certain conditions

Example: “If the weather was nice, we always went to the beach.”

2.6 Alternative Conditional Structures

Unless (meaning “if not”):

- “Unless you study, you will not pass.” (= If you don’t study, you will not pass)

As long as (meaning “on the condition that”):

- “As long as you maintain a 3.5 GPA, you will keep your scholarship.”

Provided that (meaning “only if”):

- “Provided that all requirements are met, your application will be approved.”

3 Conjunctions and Linking Words

Conjunctions are the glue that connects ideas. Strong writers use varied and appropriate conjunctions.

3.1 Conjunctions of Reason (Why Something Happens)

Because: The most common choice

- “He succeeded because he practiced consistently.”

As / Since: Slightly more formal

- “Since she understood the concept, she completed the project successfully.”

Due to / Owing to: Formal, often used in academic writing

- “The experiment was postponed due to equipment failure.”

Consequence Connectors: Therefore, so, consequently, as a result

- “She studied hard; therefore, she achieved excellent results.”

3.2 Conjunctions of Contrast (Showing Opposition)

Although / Though / Even though: Introduces the surprising element

- “Although the task was difficult, they persevered.”

However / But / Yet: Shows contradiction between ideas

- “He was tired; however, he continued working.”

Despite / In spite of: Formal, followed by noun phrase

- “Despite facing numerous obstacles, the team succeeded.”

3.3 Conjunctions of Addition (Adding Ideas)

And: Basic connection **Also / Furthermore / Moreover / Additionally:** Adds emphasis

Not only...but also: Emphasizes both points

- “Not only did she excel academically, but also she showed leadership.”

Either...or / Neither...nor: Presenting alternatives

- “Either you submit the work on time, or you face penalties.”
- “Neither option is appealing to our team.”

3.4 Conjunctions of Purpose (Why Something Is Done)

So that / In order that: To achieve an intended result

- “He studied extensively so that he could excel on the exam.”

To / In order to: Simpler, more direct

- “She attended the seminar to enhance her knowledge.”

3.5 Conjunctions of Comparison

As...as / The same as: Similarity

- “His approach is as effective as hers.”

Unlike / Rather than: Contrast in comparison

- “Unlike traditional methods, this approach is more efficient.”

Chapter IV

Word Mastery and Vocabulary Development

1 Compound Words and Word Formation

Compound words are combinations of two or more words that create a new meaning. Understanding compound words expands your vocabulary significantly.

1.1 Three Types of Compound Words

1. Open Compound Words (Spaces Between)

These are written as separate words but function as a single concept.

Examples:

- high school
- post office
- test drive
- time management
- data analysis
- field research

Pluralization: Add -s to the main noun: “high schools”, “test drives”

2. Closed Compound Words (No Spaces)

These are written as single words.

Examples:

- airport (air + port)
- background (back + ground)
- breakthrough (break + through)
- laboratory (labor + atory)
- notebook (note + book)
- textbook (text + book)

- classroom (class + room)
- homework (home + work)

3. Hyphenated Compound Words

These use hyphens to connect words.

Examples:

- mother-in-law (plural: mothers-in-law)
- twenty-year-old (describing age)
- up-to-date (current)
- state-of-the-art (most advanced)
- well-known (famous)
- mind-blowing (amazing)

1.2 Creating Compound Words

Many new words are created by combining existing words:

Noun + Noun:

- science + fiction = science fiction
- data + base = database

Adjective + Noun:

- high + school = high school
- short + cut = shortcut

Verb + Adverb:

- break + through = breakthrough
- set + up = setup

2 Idioms, Expressions, and Figurative Language

Idioms are phrases where the meaning cannot be understood from the individual words. Mastering idioms is essential for sounding natural in English.

2.1 20 Essential STEM-Appropriate Idioms

1. **Break the ice:** Start a conversation or make people comfortable
 - “During the first lab session, the teacher broke the ice with an engaging experiment.”
2. **Piece of cake:** Something very easy
 - “Once you understand the concept, the practice problems are a piece of cake.”
3. **Hit the nail on the head:** Get something exactly right
 - “Your analysis hit the nail on the head.”
4. **Costs an arm and a leg:** Very expensive
 - “That laboratory equipment costs an arm and a leg.”
5. **Cry over spilt milk:** Complain about past mistakes you can’t change
 - “Don’t cry over spilt milk about yesterday’s test; focus on the next one.”
6. **Better late than never:** Doing something late is better than not doing it
 - “Better late than never, but next time submit on time.”
7. **Under the weather:** Feeling sick
 - “I was under the weather, so I missed the lecture.”
8. **Cold feet:** Becoming nervous before an important event
 - “I had cold feet before the presentation, but it went well.”
9. **Green with envy:** Very jealous
 - “My classmates were green with envy when I won the science award.”
10. **Out of the blue:** Unexpected
 - “Out of the blue, I discovered a shortcut in the algorithm.”
11. **Give someone the cold shoulder:** Deliberately ignore someone
 - “Don’t give your study partner the cold shoulder.”
12. **Bite the bullet:** Endure something difficult
 - “I had to bite the bullet and retake the exam.”
13. **Over the moon:** Extremely happy
 - “I was over the moon when I got accepted to the university.”
14. **Kill two birds with one stone:** Accomplish two things with one action
 - “By presenting the research, I killed two birds with one stone: fulfilled the requirement and impressed the committee.”

15. **Cut corners:** Do something quickly or cheaply, sacrificing quality
 - “Never cut corners in scientific research.”
16. **Call it a day:** Stop working, finish for the day
 - “We’ve made good progress; let’s call it a day.”
17. **Drop in the ocean:** A very small amount
 - “100 is a drop in the ocean compared to the research budget.”
18. **Wrap your head around:** Understand
 - “I finally wrapped my head around quantum mechanics.”
19. **Breakthrough:** Important discovery (often used literally in STEM)
 - “The team made a major breakthrough in cancer research.”
20. **On the same page:** In agreement about something
 - “We need to be on the same page about the project methodology.”

3 Antonyms, Synonyms, and Homonyms

Building a strong vocabulary involves understanding word relationships.

3.1 Antonyms (Opposite Meanings)

Word	Antonym
achieve	fail
accelerate	decelerate
analyze	synthesize
ancient	modern
brave	cowardly
brilliant	dim
bright	dark
careful	careless
clean	dirty
clear	unclear
correct	incorrect
create	destroy
detailed	vague
difficult	easy
efficient	inefficient
expand	contract
explain	confuse
frequent	rare
generous	stingy
growth	decline
include	exclude
increase	decrease
innovation	tradition
intelligent	stupid
internal	external
light	heavy
liquid	solid
loose	tight
major	minor
natural	artificial
negative	positive
never	always
open	closed
optimize	compromise
partial	complete
peak	valley
precise	imprecise
profit	loss
quality	quantity
question	answer
rapid	slow
react	act
real	imaginary
relevant	irrelevant
repair	break
replace	keep
resolve	complicate
rough	smooth
safe	dangerous
shallow	deep
significant	insignificant
simple	complex
source	destination

3.2 Synonyms (Same or Similar Meanings)

Word	Synonym
ability	skill
analyze	examine
answer	response
approximately	roughly
argue	debate
assertion	claim
assume	suppose
atmosphere	mood
attempt	try
attention	focus
attractive	appealing
awful	terrible
background	context
barely	hardly
base	foundation
beginning	start
belief	conviction
benefit	advantage
big	large
brief	short
bright	intelligent
brilliant	exceptional
building	structure
buy	purchase
calm	peaceful
capability	capacity
careful	cautious
cause	reason
change	transformation
characteristic	trait
choice	option
choose	select
circumstance	situation
claim	assert
clear	obvious
close	near
closure	conclusion
coherent	logical
collaborate	cooperate
common	widespread
company	organization
complete	finish
complex	complicated
component	part
concept	idea
concern	worry
conclusion	summary
concrete	specific
conduct	carry out
confirm	verify
consequence	result
consider	think about
consist	comprise

3.3 Homonyms (Same Sound/Spelling, Different Meanings)

1. **Atmosphere:** The gases surrounding Earth / The mood of a place
 - “The atmosphere of the laboratory was conducive to research.”
 - “Scientists study the atmosphere using satellites.”
2. **Bank:** A financial institution / The side of a river
 - “I deposited money in the bank.”
 - “The river bank was lined with trees.”
3. **Beat:** To overcome / A rhythm
 - “We will beat our competitors.”
 - “Listen to the beat of the music.”
4. **Bore:** To drill / A dull person
 - “They will bore a hole through the rock.”
 - “Don’t be a bore at the party.”
5. **Bright:** Emitting light / Intelligent
 - “The bright sun illuminated the laboratory.”
 - “She is a bright student.”
6. **Can:** Metal container / Ability
 - “Put the chemicals in a can.”
 - “You can solve this problem.”
7. **Charge:** Electrical property / Cost
 - “The positive charge repels negative charges.”
 - “There is no charge for attendance.”
8. **Current:** Flow of electricity / Present time
 - “The electric current is strong.”
 - “Current technologies are advancing rapidly.”
9. **Conduct:** Behavior / To lead
 - “His conduct was professional.”
 - “Please conduct the experiment carefully.”
10. **Describe:** Portray / Definition
 - “Describe what you observed.”
 - “The description matched the results.”
11. **Dive:** To jump into water / A disreputable place

- “She will dive into the research project.”
- “That restaurant is a real dive.”

12. **Experiment:** Scientific test / Unusual attempt

- “Conduct the experiment carefully.”
- “They will experiment with new approaches.”

13. **Fine:** Of high quality / Penalty fee

- “This is fine work.”
- “The fine was 100.”

14. **Fit:** Appropriate / Physical condition

- “The solution fit the problem perfectly.”
- “She is in good physical fit.”

15. **Form:** Shape / Document

- “The chemical formed crystals.”
- “Complete the form with your information.”

16. **Ground:** Soil / Basis

- “The ground was hard.”
- “Your argument lacks ground.”

17. **Hard:** Solid / Difficult

- “This is hard metal.”
- “The exam was hard.”

18. **Head:** Top of body / Leader

- “She turned her head.”
- “He is the head of the department.”

19. **Iron:** Metal element / To press clothes

- “Iron is element 26.”
- “Iron your shirt before the presentation.”

20. **Lie:** Untruth / To rest

- “Don’t lie about your results.”
- “Lie down if you feel tired.”

Chapter V

Sentence Enhancement Techniques

1 Relative Clauses and Complex Sentences

Relative clauses add detail and sophistication to your writing by modifying nouns with additional information.

1.1 Understanding Relative Clauses

A relative clause is a dependent clause that modifies a noun. It begins with a relative pronoun (who, which, that, whose, where, when).

Basic Structure: Main clause + relative pronoun + relative clause

Example: “The student *who studied diligently* received the highest grade.”

1.2 Types of Relative Clauses

1. Defining (Restrictive) Clauses

These clauses are **ESSENTIAL** to the sentence’s meaning. Without them, the sentence is incomplete or ambiguous.

- “The experiment *that produced unexpected results* led to a breakthrough.”
- The clause is essential because it specifies **WHICH** experiment.
- No commas around the clause.

2. Non-Defining (Non-Restrictive) Clauses

These clauses provide **ADDITIONAL** information but are not essential. They are separated by commas.

- “The experiment, *which produced unexpected results*, led to a breakthrough.”
- The clause adds extra information but is not essential.
- Commas surround the clause.

1.3 Relative Pronouns and Their Uses

1. **Who** (for people as subject):
 - “The scientist *who discovered the formula* won the Nobel Prize.”
2. **Whom** (for people as object):
 - “The professor *whom I consulted* provided valuable guidance.”
3. **Which** (for things):
 - “The data, *which we collected carefully*, supported our hypothesis.”
4. **That** (for things, can be restrictive or non-restrictive):
 - “The algorithm *that we developed* is highly efficient.”
5. **Whose** (showing possession):
 - “The researcher *whose work I admire* is visiting our school.”
6. **Where** (for places):
 - “The laboratory, *where the experiments took place*, is state-of-the-art.”
7. **When** (for time):
 - “The year *when they made the discovery* changed scientific history.”

2 Countable and Uncountable Nouns

Understanding countability is essential for correct article and quantifier usage.

2.1 Countable Nouns

Countable nouns refer to individual, separate items that can be counted with numbers.

Characteristics:

- Can be preceded by a/an in singular form
- Can be made plural with -s/-es
- Can be counted: one, two, three, etc.
- Can be used with: many, few, several, a couple of

Examples:

- Student, molecule, equation, experiment, theory, hypothesis
- “I have three textbooks.”
- “Many students attended the seminar.”

2.2 Uncountable Nouns

Uncountable nouns refer to substances, materials, concepts, or abstract ideas that cannot be counted as individual units.

Characteristics:

- Cannot be preceded by a/an
- Do not have plural forms
- Cannot be directly counted
- Require quantifiers: some, much, a lot of, a great deal of

Common Uncountable Nouns:

- **Liquids:** water, oxygen, hydrogen, acid, solution
- **Solids:** sand, iron, steel, gold, copper
- **Gases:** air, nitrogen, carbon dioxide
- **Abstract Concepts:** knowledge, information, research, data, evidence, research
- **Materials:** wood, plastic, paper, glass
- **Subjects:** physics, chemistry, mathematics, biology

Examples:

- “I need *a lot of* water for this experiment.”
- “Much research has been conducted.”
- “The equipment requires *some* maintenance.”

2.3 Measuring Uncountable Nouns

When you need to count uncountable nouns, use a “quantifier + of” structure:

- a cup of water
- a liter of oxygen
- a piece of equipment
- a sheet of paper
- a bar of soap
- a grain of sand
- a drop of acid
- a specimen of rock

Example: “Add three drops of acid to the solution.”

3 Phrasal Verbs: Essential for Natural English

Phrasal verbs are combinations of verbs with prepositions or adverbs that create new meanings. They are extremely common in spoken and written English.

3.1 Understanding Phrasal Verb Structure

A phrasal verb consists of:

- A base verb (come, go, put, take, etc.)
- One or more particles (up, down, out, in, etc.)
- Together they create a new meaning: “put up with” = tolerate

3.2 70+ Essential Phrasal Verbs for STEM Students

Alphabetical Section A-D:

1. **Add up to** - Amount to; equal
 - “All the evidence adds up to only one conclusion.”
2. **Back up** - Support; create a copy
 - “Always back up your research files.”
3. **Beat down** - Reduce (price); overcome
 - “We beat down the cost of equipment.”
4. **Blow up** - Explode; enlarge
 - “The experiment blew up due to a chemical reaction.”
5. **Break down** - Stop working; analyze
 - “The equipment broke down.”
 - “Let me break down the process into steps.”
6. **Break through** - Make a sudden discovery
 - “Scientists broke through in quantum research.”
7. **Break up** - Separate; end
 - “The group broke up after the project.”
8. **Bring about** - Cause something to happen
 - “Technology brings about significant changes.”
9. **Bring in** - Introduce; earn
 - “Bring in fresh perspectives to the research.”

10. **Bring up** - Raise; mention
 - “Can you bring up the data from last week?”
11. **Burn out** - Become exhausted
 - “Don’t burn out from overworking.”
12. **Call off** - Cancel
 - “The experiment was called off due to equipment failure.”
13. **Carry on** - Continue
 - “Carry on with the research despite setbacks.”
14. **Carry out** - Execute; perform
 - “Carry out the experiment exactly as described.”
15. **Catch up** - Reach the same level; meet up
 - “You need to catch up in mathematics.”
16. **Check in** - Arrive; register
 - “Check in to your accommodation.”
17. **Check out** - Examine; leave
 - “Check out the new laboratory equipment.”
18. **Cheer up** - Become happier
 - “Cheer up! The results will improve next time.”
19. **Come across** - Encounter; seem
 - “I came across an interesting research paper.”
20. **Come back** - Return
 - “Come back to the main point.”
21. **Come down to** - Reduce to; depend on
 - “Success comes down to hard work.”
22. **Come up with** - Create; devise
 - “Come up with a hypothesis.”
23. **Concentrate on** - Focus on
 - “Concentrate on understanding the theory.”
24. **Count on** - Depend on; trust
 - “You can count on my support.”

25. **Cope with** - Deal with

- “How will you cope with the workload?”

26. **Crack down on** - Suppress; enforce rules

- “The school is cracking down on academic dishonesty.”

27. **Cut back** - Reduce

- “We must cut back on unnecessary expenses.”

28. **Cut down on** - Reduce consumption of

- “Cut down on energy usage.”

29. **Cut off** - Disconnect; interrupt

- “The power was cut off.”

30. **Deal with** - Handle; manage

- “Deal with problems logically.”

31. **Depend on** - Rely on; be determined by

- “Results depend on accurate measurements.”

32. **Dig into** - Investigate thoroughly

- “Dig into the data more carefully.”

33. **Dress up** - Wear formal clothes

- “You should dress up for the presentation.”

34. **Drop out** - Leave; quit

- “Don’t drop out of school.”

Alphabetical Section E-Z:

34. **Fill in** - Complete (a form); provide information

- “Fill in the missing data in the table.”

35. **Figure out** - Understand; solve

- “Figure out the solution to the problem.”

36. **Find out** - Discover; learn

- “Find out the results of the experiment.”

37. **Follow up** - Continue; investigate further

- “Follow up on your research findings.”

38. **Get along** - Have a good relationship

- “I get along well with my lab partner.”
39. **Get around** - Avoid; find ways
- “We’ll get around that limitation.”
40. **Get away with** - Escape punishment
- “Don’t get away with incomplete work.”
41. **Get behind** - Support; understand fully
- “I really get behind this approach.”
42. **Get into** - Enter; become interested
- “Get into the details of the theory.”
43. **Get over** - Overcome; recover from
- “Get over your fear of presentations.”
44. **Get through** - Complete; reach
- “Get through the difficult chapter.”
45. **Get up** - Wake up; stand up
- “Get up early for the laboratory session.”
46. **Give in** - Surrender; submit
- “Give in your assignment on time.”
47. **Give off** - Emit; produce
- “The chemical gives off toxic fumes.”
48. **Give up** - Quit; surrender
- “Never give up on your goals.”
49. **Go around** - Avoid; spread
- “Go around the obstacles in your research.”
50. **Go back to** - Return to
- “Go back to the beginning of the analysis.”
51. **Go forward with** - Proceed with
- “We will go forward with the project.”
52. **Go into** - Enter; examine thoroughly
- “Let’s not go into too much detail.”
53. **Go on** - Continue; proceed

- “Go on reading the research paper.”
54. **Go out** - Be extinguished; end
- “The lights went out.”
55. **Go over** - Review; examine
- “Go over your work before submission.”
56. **Go through** - Experience; examine carefully
- “Go through the steps methodically.”
57. **Go with** - Accept; choose
- “Go with the scientific evidence.”
58. **Grow up** - Mature; develop
- “Grow up and accept responsibility.”
59. **Hang on** - Wait; hold
- “Hang on while I get the results.”
60. **Hang out** - Spend time with
- “Don’t hang out in the laboratory during class.”
61. **Hang up** - Disconnect phone
- “Don’t hang up the phone.”
62. **Hold on** - Wait; persist
- “Hold on, let me check the data.”
63. **Hold up** - Delay; support
- “Hold up the equipment carefully.”
64. **Keep away from** - Avoid
- “Keep away from dangerous chemicals.”
65. **Keep on** - Continue
- “Keep on researching until you find answers.”
66. **Keep up** - Maintain pace; match
- “Keep up with the rapid pace of the course.”
67. **Keep up with** - Stay informed about
- “Keep up with the latest scientific developments.”
68. **Kick off** - Start; begin

- “The seminar kicks off at 9 AM.”
69. **Knock down** - Lower (price); demolish
- “They knocked down the price significantly.”
70. **Knock out** - Eliminate; make unconscious
- “This discovery knocked out the old theory.”
71. **Know of** - Be aware of
- “I know of several similar studies.”
72. **Look after** - Take care of
- “Look after the laboratory equipment.”
73. **Look at** - Examine; consider
- “Look at the data carefully.”
74. **Look back on** - Remember; reflect
- “Look back on your progress with pride.”
75. **Look down on** - Despise; consider inferior
- “Don’t look down on different approaches.”
76. **Look for** - Search for
- “Look for patterns in the research.”
77. **Look forward to** - Anticipate with pleasure
- “I look forward to the laboratory session.”
78. **Look in** - Check inside
- “Look in the cabinet for the equipment.”
79. **Look into** - Investigate
- “Look into the source of the error.”
80. **Look on** - Watch; observe
- “Look on as the experiment progresses.”
81. **Look out** - Be careful; watch
- “Look out for potential hazards.”
82. **Look over** - Review; examine
- “Look over your calculations before submitting.”
83. **Look up** - Search information; respect

- “Look up the definition in the dictionary.”

84. **Look up to** - Respect; admire

- “I look up to brilliant scientists.”

85. **Make into** - Transform

- “Make it into a successful project.”

86. **Make up** - Compose; invent

- “Don’t make up data.”

87. **Make up for** - Compensate

- “Make up for lost time.”

88. **Pass away** - Die

- “The great scientist passed away.”

89. **Pass on** - Convey; die

- “Pass on your knowledge to others.”

90. **Pass out** - Faint; distribute

- “Pass out the laboratory instructions.”

91. **Pass over** - Overlook; skip

- “Don’t pass over important details.”

92. **Pick out** - Select; identify

- “Pick out the correct answer.”

93. **Pick up** - Collect; learn

- “Pick up your assignment from the office.”

94. **Point out** - Indicate; mention

- “Point out the errors in the calculation.”

95. **Pull down** - Demolish; lower

- “Pull down outdated theories.”

96. **Pull off** - Accomplish; manage

- “Pull off an excellent presentation.”

97. **Pull out** - Remove; withdraw

- “Pull out your notes from class.”

98. **Pull through** - Recover; manage

- “Pull through despite difficulties.”
99. **Pull up** - Stop; raise
- “Pull up your grades this semester.”
100. **Put aside** - Set apart; save
- “Put aside your prejudices.”
101. **Put away** - Store; save
- “Put away the equipment after use.”
102. **Put back** - Return
- “Put back the file in the cabinet.”
103. **Put forward** - Propose; suggest
- “Put forward a new hypothesis.”
104. **Put in** - Insert; install
- “Put in the data accurately.”
105. **Put off** - Postpone; dislike
- “Don’t put off your studies.”
106. **Put on** - Wear; assume
- “Put on your safety goggles.”
107. **Put out** - Extinguish; produce
- “Put out the results in a clear format.”
108. **Put through** - Endure; conduct
- “Put through rigorous testing.”
109. **Put together** - Assemble; organize
- “Put together a comprehensive report.”
110. **Put up** - Construct; tolerate
- “Put up with incomplete explanations.”
111. **Read out** - Read aloud
- “Read out the findings.”
112. **Read over** - Review by reading
- “Read over your essay for errors.”
113. **Run into** - Meet by chance; encounter

- “I ran into an interesting problem.”
114. **Run through** - Review quickly; practice
- “Run through the procedure once more.”
115. **Set aside** - Reserve; disregard
- “Set aside your biases.”
116. **Set back** - Delay; cost
- “The error set back our progress.”
117. **Set down** - Write; establish
- “Set down the rules clearly.”
118. **Set off** - Start; trigger
- “Set off on the research project.”
119. **Set out** - Begin; arrange
- “Set out your objectives clearly.”
120. **Set up** - Establish; arrange
- “Set up the laboratory equipment.”
121. **Show off** - Display proudly; boast
- “Don’t show off your knowledge.”
122. **Show up** - Appear; arrive
- “Show up on time for the lecture.”
123. **Sign up** - Enroll; register
- “Sign up for the advanced course.”
124. **Sort out** - Solve; organize
- “Sort out the data logically.”
125. **Speed up** - Accelerate; go faster
- “Speed up the research process.”
126. **Spell out** - Explain clearly; write individually
- “Spell out your assumptions.”
127. **Start over** - Begin again
- “Start over with a fresh perspective.”
128. **Stick to** - Adhere to; remain faithful

- “Stick to the scientific method.”
129. **Stand out** - Be obvious; be conspicuous
- “This research stands out from others.”
130. **Stand up** - Resist; endure
- “Stand up for scientific integrity.”
131. **Step back** - Reconsider; withdraw
- “Step back and think logically.”
132. **Step up** - Increase; accelerate
- “Step up your efforts.”
133. **Straighten out** - Clarify; correct
- “Straighten out the misunderstanding.”
134. **Stretch out** - Extend; lie out
- “Stretch out the discussion.”
135. **Strike out** - Remove; fail
- “Strike out the errors.”
136. **Study up** - Learn intensively
- “Study up on quantum physics.”
137. **Submit to** - Yield to; accept
- “Submit to peer review.”
138. **Succeed in** - Accomplish
- “Succeed in your academic goals.”
139. **Sum up** - Summarize
- “Sum up your findings.”
140. **Supplement with** - Add to; complete
- “Supplement your research with additional data.”
141. **Switch to** - Change to
- “Switch to a better approach.”
142. **Take after** - Resemble; follow example
- “Take after successful scientists.”
143. **Take apart** - Disassemble

- “Take apart the equipment carefully.”
144. **Take away** - Remove; eliminate
- “The key takeaway from this research...”
145. **Take back** - Return; withdraw statement
- “Take back your incorrect conclusion.”
146. **Take in** - Understand; comprehend
- “Take in all the information.”
147. **Take off** - Remove; depart
- “Take off your safety goggles.”
148. **Take on** - Accept; undertake
- “Take on challenging projects.”
149. **Take out** - Remove; extract
- “Take out the key points.”
150. **Take over** - Assume control
- “Take over the project leadership.”
151. **Take up** - Begin; occupy
- “Take up a new research area.”
152. **Tear apart** - Criticize harshly; dismantle
- “Tear apart the weak arguments.”
153. **Tear down** - Destroy; criticize
- “Tear down false theories.”
154. **Thank for** - Express gratitude
- “Thank you for your contribution.”
155. **Think about** - Consider
- “Think about the implications.”
156. **Think of** - Recall; consider
- “Think of other approaches.”
157. **Think over** - Consider carefully
- “Think over the options.”
158. **Think through** - Plan; consider thoroughly

- “Think through the methodology.”

159. **Throw away** - Discard

- “Throw away irrelevant data.”

160. **Throw out** - Reject; discard

- “Throw out the contaminated sample.”

161. **Throw up** - Vomit; construct quickly

- “Throw up some ideas for discussion.”

162. **Tie up** - Complete; occupy

- “Tie up loose ends.”

163. **Tire out** - Exhaust

- “Don’t tire out yourself with overwork.”

164. **Touch on** - Mention briefly

- “Touch on the main points.”

165. **Trade off** - Exchange

- “Trade off accuracy for speed.”

166. **Track down** - Find; locate

- “Track down the source of the error.”

167. **Trade in** - Exchange for credit

- “Trade in old equipment.”

168. **Transform into** - Change completely

- “Transform into a successful researcher.”

169. **Try out** - Test; attempt

- “Try out the new procedure.”

170. **Turn around** - Reverse; improve

- “Turn around your academic performance.”

171. **Turn away** - Reject; refuse entry

- “Turn away from incorrect methods.”

172. **Turn back** - Return; reverse

- “Turn back from the wrong path.”

173. **Turn down** - Reject; lower

- “Turn down the volume.”
174. **Turn in** - Submit; deliver
- “Turn in your project on time.”
175. **Turn into** - Transform to
- “Turn ideas into reality.”
176. **Turn off** - Switch off; dislike
- “Turn off the equipment after use.”
177. **Turn on** - Switch on; become hostile
- “Turn on the analytical skills.”
178. **Turn out** - Result; produce
- “Turn out exceptional research.”
179. **Turn over** - Flip; transfer
- “Turn over the page.”
180. **Turn to** - Seek help from; change to
- “Turn to your mentor for guidance.”
181. **Turn up** - Increase; appear
- “Turn up the intensity of effort.”
182. **Understand of** - Comprehend
- “Develop a better understanding of the topic.”
183. **Use up** - Consume completely
- “Use up all the reagents.”
184. **Used to** - Habit in the past
- “I used to struggle with mathematics.”
185. **Warm up** - Heat; prepare
- “Warm up before exercise.”
186. **Wash out** - Cleanse; fail
- “Wash out the containers thoroughly.”
187. **Watch out** - Be careful
- “Watch out for potential hazards.”
188. **Water down** - Weaken; dilute

- “Water down the harsh criticism.”
189. **Wear down** - Exhaust; erode
- “Don’t let criticism wear you down.”
190. **Wear off** - Diminish; fade
- “The effect will wear off.”
191. **Wear out** - Exhaust; become unusable
- “Wear out the equipment with careful use.”
192. **Weigh down** - Burden; depress
- “Don’t let stress weigh you down.”
193. **Weigh up** - Consider; evaluate
- “Weigh up the pros and cons.”
194. **Welcome** - Greet warmly
- “Welcome to the laboratory.”
195. **Wipe out** - Destroy; eliminate
- “Wipe out all contamination.”
196. **Withdraw from** - Leave; retreat
- “Withdraw from a bad situation.”
197. **Work on** - Focus on; improve
- “Work on your presentation skills.”
198. **Work out** - Exercise; solve
- “Work out the mathematical problem.”
199. **Work through** - Process; solve
- “Work through the difficulties systematically.”
200. **Work up** - Develop; create
- “Work up an appetite for learning.”
201. **Wrap up** - Conclude; finish
- “Wrap up the project.”
202. **Write down** - Record; note
- “Write down your observations.”
203. **Write up** - Prepare a report

- “Write up your findings professionally.”

204. **Yield to** - Submit to; concede

- “Yield to scientific evidence.”

205. **Zero in on** - Focus on; target

- “Zero in on the main issue.”

Chapter VI

Advanced Language Mastery

1 Prefixes and Suffixes: Building Your Word Power

Understanding prefixes and suffixes allows you to decode unknown words and create variations of familiar words. This significantly accelerates vocabulary acquisition.

1.1 Prefixes (Added to the Beginning)

Prefix	Meaning	Examples	Application
a-, an-	without	abiotic, anhydrous, amoral	Negation
anti-	against	antibiotic, antisocial, antithesis	Opposition
auto-	self	automatic, autobiography, autonomous	Self
bi-	two	bilateral, biannual, bisect	Pair/Double
bio-	life	biography, biology, biosphere	Life science
co-	together	cooperate, coexist, collaborate	Union
counter-	opposite	counterargument, counteract	Opposition
de-	reverse/reduce	decrease, decode, defrost	Reversal
dis-	not/opposite	disappear, disagree, dishonest	Negation
dys-	bad/difficult	dysfunction, dyslexia, dysphoria	Difficulty
ex-	out/former	exit, extract, ex-president	Removal
extra-	beyond	extraordinary, extracurricular	Beyond
geo-	earth	geography, geology, geometry	Earth
hyper-	excessive	hyperactive, hypertension	Excess
hypo-	under	hypothetical, hypotenuse, hypocrisy	Below
il-, im-, in-, ir-	not	illegal, impossible, incorrect, irresponsible	Negation
inter-	between	international, interpret, interaction	Between
intra-	within	intravenous, intramural	Within
iso-	equal	isosceles, isotope, isometric	Equal
macro-	large	macroscope, macroeconomics	Large
micro-	small	microscope, microbiology	Small
mis-	wrong	misunderstand, misbehave, misjudge	Error
mono-	one	monopoly, monolingual, monotone	Single
multi-	many	multilingual, multimedia, multifaceted	Multiple
neo-	new	neoconservative, neoclassical	New
non-	not	nonfiction, nonprofit, nonsense	Negation
over-	excessive	overwork, overcrowded, oversight	Excess
para-	beside/beyond	paraphrase, paradox, parapsychology	Beyond
peri-	around	perimeter, peripheral, peritonitis	Around
poly-	many	polygon, polyglot, polynomial	Many
post-	after	postwar, postpone, postmodern	After
pre-	before	prevent, prefix, prehistoric	Before
pro-	forward/for	proceed, promote, professional	Forward
pseudo-	false	pseudoscience, pseudonym	False
quad-	four	quadrilateral, quadrant, quadruplets	Four
re-	again	repeat, reread, recycle, re-	Repetition

1.2 Suffixes (Added to the End)

Noun Suffixes (Creating Nouns)

Suffix	Meaning/Use	Examples
-tion/-sion	Action/Result	creation, interpretation, decision, conclusion
-ment	State/Result/Object	achievement, equipment, treatment, movement
-ness	Quality/State	happiness, intelligence, kindness, darkness
-ity/-ty	Quality/Condition	clarity, security, university, activity
-ance/-ence	State/Quality	importance, existence, attendance, difference
-ship	Relationship/Office	friendship, leadership, membership, scholarship
-hood	Condition/Group	childhood, adulthood, brotherhood, neighborhood
-dom	Domain/Condition	kingdom, wisdom, freedom, random
-age	Collective/Result	package, language, storage, advantage
-ism	Ideology/Practice	biology, socialism, realism, optimism
-ist	Person who practices	scientist, artist, specialist, journalist
-ity	Quality	ability, possibility, creativity, productivity
-or/-er	Person who does	teacher, actor, scientist, counselor

Adjective Suffixes (Creating Adjectives)

Suffix	Meaning/Use	Examples
-able/-ible	Capable of being	comfortable, responsible, visible, edible
-al	Relating to	personal, general, technical, musical
-ful	Full of	helpful, beautiful, powerful, meaningful
-less	Without	hopeless, endless, thoughtless, homeless
-ous/-ious	Full of/Characterized by	famous, dangerous, tedious, curious
-ive	Tending to	creative, active, attractive, impressive
-ic	Relating to	scientific, dramatic, academic, historic
-ish	Somewhat/Like	foolish, childish, reddish, selfish
-ent/-ant	Tendency/Agent	different, important, independent, relevant
-ary	Relating to	necessary, voluntary, secondary, primary

Adverb Suffixes (Creating Adverbs)

Suffix	Meaning/Use	Examples
-ly	Manner/Frequency	quickly, carefully, frequently, obviously
-ward/-wards	Direction	forward, backward, downward, upward
-wise	Direction/Manner	clockwise, counterclockwise, likewise

1.3 Word Families: Understanding Related Words

Many words are related through common roots and affixes. Understanding word families helps you learn multiple words efficiently.

Example Word Family: SCIENCE

- **Science** (noun): systematic study of nature
- **Scientist** (noun): person who practices science
- **Scientific** (adjective): relating to science
- **Scientifically** (adverb): in a scientific manner
- **Unscientific** (adjective): not based on science
- **Pseudoscience** (noun): false science

Example Word Family: CREATE

- **Create** (verb): make something new
- **Creator** (noun): person who creates
- **Creation** (noun): something created
- **Creative** (adjective): having creative ability
- **Creativity** (noun): quality of being creative
- **Recreate** (verb): create again
- **Procreate** (verb): produce offspring

2 Special Usage and Common Distinctions

2.1 Make vs. Do: A Complete Analysis

These two verbs cause frequent confusion because they can have overlapping meanings. Understanding the distinction will significantly improve your writing.

USE MAKE WHEN:

1. Creating or producing something tangible:

- “Make a model of the molecular structure.”
- “Make a cup of tea.”
- “Make a photocopy.”

2. Creating abstract things:

- “Make a decision.”
- “Make progress.”
- “Make a difference.”
- “Make an excuse.”

3. Achieving financial results:

- “Make money.”
- “Make a profit.”
- “Make sales.”

4. Causing someone to be in a state:

- “Make someone happy.”
- “Make someone nervous.”
- “Make someone tired.”

5. Common Make Collocations:

- Make an appointment, attempt, attack, bed, call, comment, complaint, effort, error, excuse, face, film, list, meal, mess, mistake, note, offer, phone call, plan, point, promise, request, reservation, sacrifice, sense, shot, sound, speech, suggestion, threat

USE DO WHEN:**1. Performing actions or activities:**

- “Do the homework.”
- “Do exercise.”
- “Do the washing.”
- “Do research.”

2. Completing work or tasks:

- “Do a good job.”
- “Do your best.”
- “Do the cooking.”
- “Do business.”

3. Professional/Academic work:

- “Do an experiment.”
- “Do a presentation.”
- “Do a project.”
- “Do research.”

4. General actions (less specific):

- “What do you do?” (What is your job?)
- “Do something about it.”
- “Do nothing.”

5. Common Do Collocations:

- Do an exam, business, calculations, chores, cooking, damage, dishes, duty, exercise, experiment, favor, good, harm, homework, housework, laundry, nothing, operation, report, research, something, work

Quick Memory Tool:

MAKE = Create/Produce something (usually with a result you can see or measure)

DO = Perform an action or activity (usually without creating a tangible object)

Think of it this way: You *make* a cake, but you *do* the cooking.

2.2 On Time vs. In Time

These prepositions are frequently confused, but they have distinct meanings in English.

ON TIME = At the correct or scheduled time; punctually

- “The train arrived **on time**.” (It arrived at the expected time)
- “Submit your assignment **on time**.” (By the deadline)

- “The meeting starts **on time** at 3 PM.” (At 3 PM exactly)

IN TIME = Before it is too late; with enough time to spare

- “We arrived **in time** to catch the plane.” (We made it before departure)
- “He completed the work **in time** for the deadline.” (He finished before the deadline)
- “Can you finish **in time**?” (Before it becomes too late)

Memory Aid:

ON TIME = Specific time (on schedule, on the clock)

IN TIME = Time enough (before it’s too late)

2.3 Other Important Distinctions

Wide vs. Width

Both are correct, but they are different parts of speech.

- **Wide** (adjective): “This road is very **wide**.”
- **Width** (noun): “The **width** of this road is 50 meters.”
- **Alternative**: “The road is 50 meters **in width**.”

Pandemic vs. Epidemic

Both describe disease outbreaks, but the scale differs significantly.

- **Epidemic**: A disease outbreak that spreads rapidly in a specific region or population
 - “The Ebola epidemic affected West Africa.”
- **Pandemic**: A disease outbreak that spreads across multiple countries or continents; affects the global population
 - “The COVID-19 pandemic affected the entire world.”

Advice vs. Advise

- **Advice** (noun): Recommendations or suggestions
 - “I received valuable **advice** from my mentor.”
- **Advise** (verb): To offer recommendations or suggestions
 - “My professor **advised** me to study harder.”

Affect vs. Effect

- **Affect** (verb): To influence or change
 - “Climate change affects ocean temperatures.”
- **Effect** (noun): A result or consequence; (verb): to bring about
 - “The **effect** of warming is visible globally.”
 - “We need to **effect** change.” (cause change to occur)

3 Context Clues and Advanced Vocabulary

Learning to infer meaning from context is a crucial skill for reading comprehension and vocabulary development.

3.1 Types of Context Clues

1. Definition Context Clues

The text directly defines or explains an unfamiliar word.

Example: “The organism was bioluminescent, meaning it could produce its own light.”

2. Synonym Context Clues

A similar word or phrase appears nearby.

Example: “The scientist’s erudite, scholarly approach to research impressed everyone.”

3. Antonym Context Clues

An opposite word or concept appears nearby.

Example: “Unlike his taciturn colleague, Dr. Smith was quite loquacious in meetings.” (You infer taciturn means quiet/reserved)

4. Explanation Context Clues

The surrounding words provide an explanation of the unknown word.

Example: “The data was incongruent; the results didn’t match our predictions at all.”

5. Example Context Clues

Specific examples clarify the meaning.

Example: “The professor mentioned several parsimonious scientists—those who were extremely careful with their money and resources.”

6. Description Context Clues

Descriptive phrases provide meaning.

Example: “The dilapidated laboratory, with its crumbling walls and broken equipment, needed complete renovation.”

7. Comparison Context Clues

Similarities help establish meaning.

Example: “Just as a catalyst speeds up a chemical reaction, a facilitator accelerates group progress.”

3.2 Advanced Vocabulary for STEM Students

Word	Definition	STEM Context
Aberration	Deviation from normal	"The unexpected result was an aberration in the data."
Abeyance	State of suspension	"The project is in abeyance pending funding."
Abjure	To re-ject/renounce	"We must abjure unscientific methods."
Abnegate	To deny/reject	"Scientists abnegate personal bias."
Abscond	To leave secretly	"Data cannot abscond from proper storage."
Absolve	To free from blame	"New evidence absolves the original hypothesis."
Abstain	To refrain from	"Abstain from contaminating the sample."
Abstruse	Difficult to understand	"The mathematics was abstruse but necessary."
Abundance	Great quantity	"The abundance of evidence supports the theory."
Accelerate	Speed up	"New technology accelerates research."
Accessible	Easily reached/understood	"Make research accessible to all."
Acclimatize	Adjust to new environment	"Acclimatize equipment to lab conditions."
Accolade	Praise/honor	"The scientist received accolades for discoveries."
Accommodate	Make suitable for	"Accommodate different learning styles."
Accompany	Go together with	"Evidence accompanies the hypothesis."
Accomplish	Complete successfully	"Accomplish your research objectives."
Accord	Agreement/harmony	"The results accord with expectations."
Accretion	Gradual growth/accumulation	"Scientific knowledge grows through accretion."

Chapter VII

Practical Application and Integrated Skills

1 Common Mistakes and How to Avoid Them

1.1 The 20 Most Common English Mistakes Among STEM Students

1. Subject-Verb Agreement Errors

Mistake: “The results shows that...”

Correct: “The results show that...” (Results is plural)

Rule: Verify that your verb agrees with the subject in number. Collective nouns (data, criteria, phenomena) often take plural verbs.

2. Confusing Its vs. It’s

Mistake: “The experiment completed it’s final phase.”

Correct: “The experiment completed its final phase.”

Rule: Its = possession; It’s = it is

3. Inconsistent Tense

Mistake: “We collected data and analyze the results...”

Correct: “We collected data and analyzed the results...”

Rule: Maintain consistent tenses within a paragraph/section unless indicating a time shift.

4. Dangling Modifiers

Mistake: “After completing the experiment, the results were analyzed.”

Correct: “After completing the experiment, we analyzed the results.”

Rule: The subject of the introductory phrase must be the same as the subject of the main clause.

5. Run-on Sentences

Mistake: “The data was significant it proved the hypothesis was correct.”

Correct: “The data was significant. It proved the hypothesis was correct.” OR “The data was significant; it proved the hypothesis was correct.”

Rule: Don’t connect two independent clauses without proper punctuation or conjunctions.

6. Sentence Fragments

Mistake: “Which was completely unexpected.” (Incomplete thought)

Correct: “The result was unexpected, which was completely surprising.”

Rule: Every sentence must have a subject and a complete verb.

7. Misusing Affect/Effect

Mistake: “This will effect the outcome.”

Correct: “This will affect the outcome.” (verb = influence)

Rule: Affect = verb (to influence); Effect = noun (result) or verb (to cause)

8. Using Double Negatives

Mistake: “The results weren’t not significant.”

Correct: “The results were not significant.” OR “The results were significant.”

Rule: One negative is sufficient in English.

9. Incorrect Article Usage

Mistake: “An university offers excellent research.” OR “The data are incomplete.”

Correct: “A university offers excellent research.” (A before consonant sound) OR “The data is incomplete.” (Data can be singular in modern English)

Rule: Use ‘a’ before consonant sounds, ‘an’ before vowel sounds. Understand whether nouns are countable or uncountable.

10. Misplaced Modifiers

Mistake: “I only can analyze the data.”

Correct: “I can only analyze the data.”

Rule: Place modifiers as close as possible to the word they modify.

11. Incorrect Pronoun Usage

Mistake: “Between you and I, the experiment failed.”

Correct: “Between you and me, the experiment failed.” (object pronoun after preposition)

Rule: Use object pronouns after prepositions.

12. Comma Splice

Mistake: “The hypothesis was tested, the results supported it.”

Correct: “The hypothesis was tested, and the results supported it.” OR “The hypothesis was tested; the results supported it.”

Rule: Don’t use a comma to connect two independent clauses.

13. Mixing Active and Passive Voice Inconsistently

Mistake: “We conducted the experiment, and the results were analyzed by us.”

Correct: “We conducted the experiment and analyzed the results.”

Rule: Maintain consistency in voice unless there’s a specific reason to switch.

14. Using Incorrect Comparative Forms

Mistake: “This method is more better than that one.”

Correct: “This method is better than that one.” OR “This method is far superior to that one.”

Rule: Use comparative form (better) OR more + adjective (more efficient), not both.

15. Incomplete Comparisons

Mistake: “Method A is more efficient.” (Efficient than what?)

Correct: “Method A is more efficient than Method B.”

Rule: Always complete comparisons with ‘than’ or an explicit second item.

16. Misusing Who vs. Whom

Mistake: “The scientist whom conducted the research...”

Correct: “The scientist who conducted the research...” (Subject = who)

Rule: Who = subject (performs action); Whom = object (receives action)

17. Incorrect Verb Forms

Mistake: “I should of known better.”

Correct: “I should have known better.”

Rule: Use ‘have’, not ‘of’, with modal verbs.

18. Misusing That vs. Which

Mistake: “The data, that was significant, proved the theory.”

Correct: “The data, which was significant, proved the theory.”

Rule: Which = non-defining clause (extra info, use commas); That = defining clause (essential info, no commas)

19. Apostrophe Errors in Plurals

Mistake: “The result’s are significant.”

Correct: “The results are significant.”

Rule: Don’t use apostrophes for plural forms; use them for possessives and contractions only.

20. Inconsistent Capitalization

Mistake: “I conducted The Experiment with my professor.”

Correct: “I conducted the experiment with my professor.”

Rule: Capitalize proper nouns, the beginning of sentences, and titles only.

2 Academic Writing Techniques

2.1 Writing Clear, Professional Sentences

Principle 1: Use Active Voice (Generally)

Active voice is stronger and clearer than passive voice.

- **Passive:** “It is believed that the hypothesis was correct.”
- **Active:** “We believe the hypothesis is correct.”

Principle 2: Be Specific and Concrete

- **Vague:** “The results were interesting.”
- **Specific:** “The results revealed a 45

Principle 3: Avoid Wordy Constructions

- **Wordy:** “It is important to note that the data indicates...”
- **Concise:** “The data indicates...”

Principle 4: Use Parallel Structure

- **Non-parallel:** “We studied, tested, and were analyzing the results.”
- **Parallel:** “We studied, tested, and analyzed the results.”

Principle 5: Maintain Subject Consistency

- **Inconsistent:** “We analyzed the data. The results showed significant patterns.”
- **Consistent:** “We analyzed the data and identified significant patterns.”

2.2 Paragraph Structure for Academic Writing

Effective paragraph structure includes:

1. **Topic Sentence:** Clearly states the main idea
 - **Example:** “Temperature significantly affects enzyme activity in biochemical reactions.”
2. **Supporting Sentences:** Provide evidence, explanation, or examples
 - Use data, research findings, or logical reasoning
 - Cite sources appropriately
3. **Analysis:** Explain the significance of the information
 - Discuss implications and connections to the main idea
4. **Concluding Sentence:** Summarizes or transitions
 - Reinforces the main idea or creates a bridge to the next paragraph

3 Effective Communication in STEM Contexts

3.1 Writing Scientific Reports

Key Characteristics:

- **Objective tone:** Maintain impartiality and professionalism
- **Precise language:** Use exact, technical terminology
- **Organized structure:** Follow standard scientific report format
- **Evidence-based:** Support all claims with data and citations
- **Clear methodology:** Explain procedures so others can replicate them

Standard Scientific Report Sections:

1. **Title:** Clear, descriptive, accurate
2. **Abstract:** 200-250 words summarizing the entire study
3. **Introduction:** Background, context, research question, hypothesis
4. **Methods:** Detailed procedures, materials, design
5. **Results:** Findings presented objectively (often with tables/figures)
6. **Discussion:** Interpretation of results, implications, limitations
7. **Conclusion:** Summary of findings and significance
8. **References:** All sources cited in standard format (APA, MLA, etc.)

3.2 Presenting Research Effectively

Oral Presentation Tips:

1. **Begin with a hook:** Start with an engaging question or statement that captures attention
2. **Clearly state your thesis:** Make your main argument immediately clear
3. **Use visuals effectively:** Slides should support, not duplicate, your speech
4. **Speak with conviction:** Demonstrate confidence in your research
5. **Practice timing:** Ensure you stay within time limits
6. **Anticipate questions:** Prepare for likely inquiries from the audience
7. **Conclude powerfully:** End with clear take-home messages

Creating Effective Slides:

- **Minimize text:** Use keywords, not full sentences
- **One idea per slide:** Don't overcrowd information

- **Use visuals:** Charts, graphs, and images enhance understanding
- **Consistent formatting:** Maintain uniform fonts and colors
- **Readable fonts:** Ensure audience can read from distance
- **Cite sources:** Give credit to others' work

4 Transitional Phrases and Connectors

Transitions help readers follow your logic and understand relationships between ideas.

4.1 Transitions by Function

Adding Information:

- additionally, furthermore, moreover, in addition, also, as well as, besides, too, furthermore

Showing Contrast:

- however, on the other hand, in contrast, nevertheless, still, yet, whereas, unlike, conversely

Showing Cause and Effect:

- therefore, consequently, as a result, thus, because of this, for this reason, as a consequence

Showing Sequence/Order:

- first, second, third, next, then, finally, subsequently, meanwhile, initially

Giving Examples:

- for example, for instance, specifically, namely, to illustrate, such as, in particular

Emphasizing:

- indeed, certainly, obviously, clearly, undoubtedly, in fact, particularly, especially

Summarizing:

- in summary, in conclusion, in short, to summarize, ultimately, in essence, in brief

Qualifying:

- perhaps, possibly, arguably, reportedly, allegedly, seemingly, to some extent, relatively

Chapter VIII

Assessment and Practice

1 Quick-Reference Summary Tables

1.1 Verb Tenses at a Glance

Tense	Structure	When to Use	Example
Simple Present	S + V	Habits, facts, schedules	I study every day.
Present Continuous	S + am/is/are + V-ing	Now/current actions	I am studying now.
Simple Past	S + V-ed	Completed past actions	I studied yesterday.
Past Continuous	S + was/were + V-ing	Past action interrupted	I was studying when he called.
Present Perfect	S + have/has + V-en	Recent or general past	I have studied hard.
Past Perfect	S + had + V-en	Before another past action	I had studied before he called.
Simple Future	S + will + V	Future predictions	I will study tomorrow.
Future Continuous	S + will + be + V-ing	Ongoing future action	I will be studying tonight.
Future Perfect	S + will + have + V-en	Before a future time	I will have studied by then.
Pres. Perf. Cont.	S + have/has + been + V-ing	Duration up to now	I have been studying for hours.

1.2 Common Prepositions

Category	Prepositions	Usage	Example
2*Time	at, in, on	at 3 PM, in July, on Monday	2*Meet at 3 PM on Friday
	before, after, during	temporal relationships	
2*Place	in, at, on	enclosed, point, surface	2*Data on the table in the lab
	under, above, between	position	
2*Direction	to, from, toward, into	movement	2*Go to the laboratory
	up, down, across	direction	

1.3 Conjunction Summary

Type	Conjunctions
Coordinating	and, but, or, nor, yet, so, for
Subordinating (Time)	when, while, as, before, after, since, until
Subordinating (Condition)	if, unless, as long as, provided that
Subordinating (Reason)	because, as, since, so that
Subordinating (Contrast)	although, though, even though, whereas
Correlative	either...or, neither...nor, both...and, not only...but also

2 Self-Assessment Checklist

Use this checklist to evaluate your English proficiency in each area:

2.1 Grammar Mastery

Verb Tenses

I can correctly form and use all 13 verb tenses

I maintain consistent tenses in writing

I understand when to use each tense appropriately

Pronouns and Agreement

Subject and object pronouns are used correctly

Possessive forms are accurate (its/it's)

Pronouns agree with antecedents in number and gender

Sentence Structure

I avoid fragments and run-on sentences

I use complex sentences with proper subordination

Parallel structure is consistent

Active and Passive Voice

I prefer active voice for clarity

I use passive voice when appropriate

I maintain consistency within paragraphs

2.2 Vocabulary and Expressions

Vocabulary Knowledge

- I understand and use 100+ academic vocabulary words
- I can identify synonyms and antonyms
- I recognize and use phrasal verbs naturally
- I understand common idioms and figurative language

Word Formation

- I understand common prefixes and suffixes
- I can infer word meanings from affixes
- I recognize word families and related forms

Context Clues

- I can determine word meanings from context
- I use context to improve comprehension

2.3 Academic Communication

Writing Skills

- I write clear, well-organized paragraphs
- I use appropriate transitions
- I maintain academic tone and formal language
- I cite sources properly

Presentation Skills

- I communicate ideas clearly in presentations
- I use visuals effectively
- I answer questions confidently

Critical Reading

- I comprehend complex academic texts
- I identify main ideas and supporting details
- I evaluate arguments critically

2.4 Common Mistakes Avoidance

- Subject-verb agreement is consistently correct
- Verb tenses are appropriate and consistent
- Pronouns agree with antecedents
- No comma splices or run-on sentences

Affect/effect, its/it's, and similar pairs are used correctly

Modifiers are placed correctly

Sentences are complete (no fragments)

Parallel structure is maintained

Contractions are appropriate for register

Punctuation is accurate throughout

3 Final Words of Encouragement

Mastering English is a continuous journey that extends throughout your life. The fact that you are studying this comprehensive guide demonstrates your commitment to excellence.

Remember these key principles as you progress:

1. Consistency Matters More Than Perfection

Small, regular practice is far more effective than infrequent intense study. Set aside 20-30 minutes daily for English learning, and you will see remarkable progress.

2. Active Practice Beats Passive Reading

Don't just read this guide—use it actively. Write sentences, speak aloud, teach concepts to others, and apply learning to real situations.

3. Learn from Mistakes

When you make errors, celebrate them as learning opportunities. Analyze what went wrong, review the rule, and practice until mastery.

4. Embrace Challenges

Read more difficult texts, listen to native speakers, and challenge yourself. Growth happens at the edge of your comfort zone.

5. Build Your Personal Language Resources

Create a vocabulary journal, maintain phrase lists, and develop example sentences. These personal resources become invaluable references.

6. Connect with English Speakers

Engage with native and non-native English speakers. Real conversations are the best classroom for authentic language learning.

7. Stay Motivated

Remember why you're learning English. Whether for academic success, career advancement, or personal enrichment, keep your ultimate goal in mind.

Your Success is Our Mission

Excellence in English opens doors to global opportunities

The STEM High School for Boys 6th of October City
STEM Mentorship Programme 2026